

Ex. Nos. 10	Wireless Network Security Audit using Aircrack-ng
Date :	

Aim :

To perform wireless network analysis and security auditing using Aircrack-ng tools in Kali Linux for educational and ethical hacking purposes.

Software Requirements :

Host Operating System: Windows / Linux / macOS

Virtualization Software: Oracle VirtualBox

Guest Operating System: Kali Linux

Security Tools: Aircrack-ng Suite

System Requirements:

- o RAM: Minimum 4 GB (8 GB recommended)
- o Disk Space: Minimum 30 GB
- o Processor: Intel/AMD with Virtualization support
- o Wireless Adapter supporting Monitor Mode

Introduction:

Wireless network security auditing involves analyzing Wi-Fi networks to identify vulnerabilities and weaknesses. The Aircrack-ng suite is a powerful set of tools used for monitoring, packet capturing, and cracking wireless network passwords.

By performing controlled auditing in a lab environment, users can understand how weak encryption and poor password practices can compromise wireless security.

Procedure:

Step 1: Start Kali Linux Virtual Machine

1. Launch Oracle VirtualBox.
2. Start the Kali Linux virtual machine.
3. Log in using Kali credentials.

Step 2: Connect Wireless Adapter

1. Attach a wireless adapter that supports monitor mode.
2. Verify adapter using:

Ex. Nos. 11	Performing IP Spoofing Using a GitHub Tool in Kali Linux (Education Demonstration)
Date :	

Aim

To demonstrate IP spoofing techniques using an open-source GitHub tool in Kali Linux for understanding network-level attack concepts and security vulnerabilities.

⚠ Important Note

This experiment is conducted strictly for academic and ethical purposes on authorized networks only.

Unauthorized IP spoofing is illegal and punishable by law.

Software Requirements

- Host Operating System: Windows / Linux / macOS
- Virtualization Software: Oracle VirtualBox
- Guest Operating System: Kali Linux
- Security Tool: John the Ripper
- System Requirements:
 - RAM: Minimum 4 GB (8 GB recommended)
 - Disk Space: Minimum 30 GB
 - Processor: Intel/AMD with Virtualization support

Introduction

IP spoofing is a technique where an attacker forges the source IP address in network packets to impersonate another system. It is commonly used in attacks such as DDoS, session hijacking, and bypassing IP-based authentication. This experiment demonstrates IP spoofing using a publicly available GitHub tool to understand how attackers manipulate packet headers. Kali Linux provides a safe and controlled environment for ethical network security experimentation.

Procedure

Step 1: Start Kali Linux Virtual Machine

1. Open Oracle VirtualBox.
2. Start Kali Linux.
3. Log in using Kali credentials.

Step 2: Open Terminal

1. Launch Terminal in Kali Linux.
2. Update system packages:
3. `sudo apt update`

Step 3: Clone IP Spoofing Tool from GitHub

1. Clone the repository:
2. `git clone https://github.com/<username>/<ip-spoofing-tool>`

3. Navigate to the tool directory:
4. `cd ip-spoofing-tool`

Step 4: Install Dependencies

1. Install required dependencies:
2. `sudo apt install python3`
3. `pip3 install -r requirements.txt`

Step 5: Configure the Tool

1. Open the script file:
2. `nano spoof.py`
3. Configure:
 - o Target IP address
 - o Spoofed source IP address
 - o Network interface

Step 6: Execute IP Spoofing

1. Run the tool with root privileges:
2. `sudo python3 spoof.py`
3. Observe packet transmission with spoofed IP addresses.

Step 7: Monitor Network Traffic (Optional)

1. Use Wireshark or tcpdump:
2. `sudo tcpdump`
3. Verify spoofed source IP packets.

Output

Packets are sent with forged source IP addresses.

Network traffic reflects spoofed IP information.

Demonstration successfully completed.

```
(student@kali)-[~]
$ git clone https://github.com/GokulkrishnaR31/NearNow.git
Cloning into 'NearNow' ...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.

(student@kali)-[~]
$ cd NearNow

(student@kali)-[~/NearNow]
$ ls
README.md

(student@kali)-[~/NearNow]
$ cat README.md
# NearNow

(student@kali)-[~/NearNow]
$
```

```
(student@kali)-[~]
$ nano spoof.py
```

```
(student@kali)-[~]
$ sudo python3 spoof.py
```

Sending spoofed packets ...

```
###[ IP ]###
version      = 4
ihl          = None
tos          = 0x0
len          = None
id           = 1
flags        =
frag         = 0
ttl          = 64
proto        = icmp
chksum       = None
src          = 8.8.8.8
dst          = 10.0.2.15
\options     \
###[ ICMP ]###
type         = echo-request
code         = 0
chksum       = None
id           = 0x0
seq          = 0x0
unused       = b''
```

```
WARNING: MAC address to reach destination not found. Using broadcast.
.WARNING: MAC address to reach destination not found. Using broadcast.
.WARNING: more MAC address to reach destination not found. Using broadcast.
.WARNING: MAC address to reach destination not found. Using broadcast.
.WARNING: MAC address to reach destination not found. Using broadcast.
```

Sent 5 packets.

Introduction

Online Clipboard is free online tool that help you

How to use:

1. choose your text or image that would be like to
2. Send your text or image to Online Clipboard by
3. At the other device, retrieve your text or image

Send to Online Clipboard:

```
09:18:06.375360 ARP, Request who-has 10.0.2.2 tell 10.0.2.15, length 28
09:18:06.376550 ARP, Reply 10.0.2.2 is-at 52:55:0a:00:02:02, length 28
```

Retrieve from Online Clipboard:

```
09:17:16.696974 IP 93.243.107.34.bc.googleusercontent.com.https > 10.0.2.15.3
9712: Flags [P.], seq 1051:1188, ack 2017, win 65535, length 137
09:17:16.747501 IP 10.0.2.15.39712 > 93.243.107.34.bc.googleusercontent.com.h
ttps: Flags [.], ack 1188, win 63053, length 0
09:17:16.096172 IP 10.0.2.15.49282 > ec2-52-43-182-82.us-west-2.compute.amazo
naws.com.https: Flags [.], ack 5521, win 65535, length 0
09:17:16.096735 IP ec2-52-43-182-82.us-west-2.compute.amazonaws.com.https > 1
0.0.2.15.49282: Flags [.], ack 2998, win 65535, length 0
09:17:26.321960 IP 10.0.2.15.49282 > ec2-52-43-182-82.us-west-2.compute.amazo
naws.com.https: Flags [.], ack 5521, win 65535, length 0
09:17:26.322143 IP ec2-52-43-182-82.us-west-2.compute.amazonaws.com.https > 1
0.0.2.15.49282: Flags [.], ack 2998, win 65535, length 0
09:17:36.545154 IP 10.0.2.15.49282 > ec2-52-43-182-82.us-west-2.compute.amazo
naws.com.https: Flags [.], ack 5521, win 65535, length 0
09:17:36.545449 IP ec2-52-43-182-82.us-west-2.compute.amazonaws.com.https > 1
0.0.2.15.49282: Flags [.], ack 2998, win 65535, length 0
09:17:46.785071 IP 10.0.2.15.49282 > ec2-52-43-182-82.us-west-2.compute.amazo
naws.com.https: Flags [.], ack 5521, win 65535, length 0
09:17:46.785542 IP ec2-52-43-182-82.us-west-2.compute.amazonaws.com.https > 1
0.0.2.15.49282: Flags [.], ack 2998, win 65535, length 0
09:17:51.903328 ARP, Request who-has 10.0.2.2 tell 10.0.2.15, length 28
09:17:51.903704 ARP, Reply 10.0.2.2 is-at 52:55:0a:00:02:02 (oui Unknown), le
ngth 50
09:17:57.025215 IP 10.0.2.15.49282 > ec2-52-43-182-82.us-west-2.compute.amazo
naws.com.https: Flags [.], ack 5521, win 65535, length 0
09:17:57.025765 IP ec2-52-43-182-82.us-west-2.compute.amazonaws.com.https > 1
0.0.2.15.49282: Flags [.], ack 2998, win 65535, length 0
09:18:07.264939 IP 10.0.2.15.49282 > ec2-52-43-182-82.us-west-2.compute.amazonaws.com.https: Flags [.], ack 5521, win 65535, length 0
09:18:07.266136 IP ec2-52-43-182-82.us-west-2.compute.amazonaws.com.https > 10.0.2.15.49282: Flags [.], ack 2998, win 65535, length 0
09:18:11.910013 IP 10.0.2.15.49282 > ec2-52-43-182-82.us-west-2.compute.amazonaws.com.https: Flags [P.], seq 2998:3022, ack 5521, win 65535, length 24
09:18:11.910071 IP 10.0.2.15.49282 > ec2-52-43-182-82.us-west-2.compute.amazonaws.com.https: Flags [F.], seq 3022, ack 5521, win 65535, length 0
09:18:11.910614 IP ec2-52-43-182-82.us-west-2.compute.amazonaws.com.https > 10.0.2.15.49282: Flags [.], ack 3022, win 65535, length 0
09:18:11.910615 IP ec2-52-43-182-82.us-west-2.compute.amazonaws.com.https > 10.0.2.15.49282: Flags [.], ack 3023, win 65535, length 0
09:18:11.913985 IP ec2-52-43-182-82.us-west-2.compute.amazonaws.com.https > 10.0.2.15.49282: Flags [F.], seq 5521, ack 3023, win 65535, length 0
09:18:11.913997 IP 10.0.2.15.49282 > ec2-52-43-182-82.us-west-2.compute.amazonaws.com.https: Flags [.], ack 5522, win 12224, length 0
```

Result

IP spoofing was successfully demonstrated using a GitHub-based tool in Kali Linux.

The experiment illustrates how attackers can manipulate IP headers and highlights the need for robust network security mechanisms.